Hamna Saleem

0451 403 325
hamna.sa@hotmail.com
linkedin.com/in/hamnasaleem

PROFESSIONAL SUMMARY & CAREER GOAL

Final-year Biomedical Engineering (Honours) student at UTS and Biomedical Engineering Technician at Royal Prince Alfred Hospital, with experience across clinical engineering, biotechnology research and medical-device development. In my current role, I perform preventative maintenance, testing, verification and fault diagnosis of medical equipment in accordance with AS/NZS 3551, working with engineering and clinical staff to support safe and reliable patient care. I combine clinical insight, scientific understanding, and electronics and design skills to translate healthcare needs into practical, patient-focused medical-device solutions. I aim to build my career in medical-device R&D, developing technologies that better meet the needs of clinicians and patients and improve healthcare quality and accessibility.

EDUCATION

Bachelor of Engineering (Honours) **Feb 2022 – 2026**
Diploma in Professional Engineering Practice
University of Technology Sydney

Major in Biomedical
- Current GPA 6.82 (High Distinction Average)
- Dean's Merit List 2025
- High Distinction in Electronics and Circuits
- Distinction in Medical Devices and Diagnostics
- High Distinction in Biomedical Instrumentation

WORK EXPERIENCE

Biomedical Electronics Technician **Jul 2026 – Present**

- Perform preventative maintenance, testing, verification and fault diagnosis of medical equipment in accordance with AS/NZS 3551, supporting equipment safety and reliability.
- Respond to equipment faults and clinician concerns across hospital departments, including the Emergency Department, troubleshooting issues and coordinating appropriate technical support.
- Manage medical equipment assets and servicing activities, coordinating with clinical staff, biomedical engineering teams and external service providers to maintain equipment availability.

Biomedical Engineering Intern **Jan 2026 – May 2026**
EnGeneIC

- Maintaining adherent and suspension mammalian cell cultures (e.g., cancer lines, CHO) to support R&D (quality control, antibody production), including thawing, seeding, and passaging.
- Completing daily culture checks (counts/viability, pH, morphology/contamination) and preparing samples for downstream workflows (e.g., transfection/ELISA support).
- Following SOPs/aseptic technique and keeping clear experimental records and handovers.

Biomedical Engineering Intern **Jan 2024 – Jul 2024**
Royal Prince Alfred and Concord Hospital, Sydney Local Health District

- Repaired and maintained medical equipment compliant with AS3551 standards, applying technical skills and analytical thinking to ensure safety and efficiency.
- Performed preventative maintenance and audits across clinical areas, utilising project management, communication, and collaboration with hospital staff to minimise disturbance.
- Coordinated with vendors and contractors to streamline the servicing and procurement of medical devices, demonstrating proactiveness and adaptability in dynamic situations.

EMPLOYMENT

Operations Team Member **Sep 2021 - Present**
Bunnings Warehouse, Bankstown Airport

- Lead and coordinate a team of 6–8 employees in a dynamic environment, ensuring operational efficiency and staff well-being.
- Manage customer inquiries through collaboration with internal teams to deliver clear, solution-focused communication.
- Adapt to changing demands to deliver high-quality customer service by applying strong problem-solving skills and product knowledge to assist in inquiries.

EXTRACURRICULAR ACTIVITIES

Undergraduate Student Representative **Mar 2025 – Jan 2026**
University of Technology Sydney, Faculty of Engineering and IT

- Collected and represented student feedback to improve inclusivity and engagement across classes and faculty activities.
- Liaised with staff and student stakeholders to communicate issues and support student-informed improvements.

Mentee **Feb 2025 – Dec 2025**
UTS Lucy Mentoring Program

- Developed wider professional networks, gained industry insights, and built strong technical and leadership skills.
- Demonstrated a commitment to succeed and a passion for learning through active participation in workshops and mentor-guided projects.

PROJECTS

StrikeDX – LFA for Rapid Differentiation of Ischemic and Haemorrhagic Strokes **2025**
Team Leader

- Developed a rapid point-of-care diagnostic using LFA to differentiate ischemic and haemorrhagic stroke within 15 minutes.
- Integrated dual biomarkers (GFAP & NT-proBNP) for improved accuracy and designed a user-friendly emergency-use cassette.
- Gained skills in biomarker selection, immunoassay development, fluorescence detection, and medical device design.

Maternix – Portable PCR Unit for Remote Field Testing of Group B Streptococcus **2025**
Research and Development Team Member

- Designed a portable point-of-care PCR diagnostic for Group B Streptococcus in pregnancy, targeting use in remote and low-resource settings outside conventional lab infrastructure.
- Defined requirements and system architecture using medical device development practices (requirements traceability, risk-aware design, verification planning).
- Developed programming skills for operation (thermal cycling control, sensing feedback, results output) and communicated outcomes via workflow diagrams and presentations.

Wireless Pulse Oximetry and Temperature Telemetry System **2025**
Research & Development Member

- Built an embedded sensing prototype to measure PPG-based pulse metrics and temperature, and stream data wirelessly for live monitoring and logging.

- Implemented signal conditioning and beat-detection logic to improve measurement reliability, then benchmarked outputs against a commercial reference device.
- Gained skills in sensor integration, embedded programming, signal processing, IoT telemetry, validation/testing, and dashboard-based data reporting**.**

TECHNICAL SKILLS

- **Electrical & Instrumentation:** Circuit troubleshooting, sensors, microcontroller-based systems, signal conditioning, verification, soldering
- **Biomedical Engineering:** Medical device servicing, equipment compliance (AS3551)
- **Programming & Data Analysis:** Java, KNIME, MATLAB
- **Design & Simulation:** SolidWorks, LabVIEW, 3D Printing
- **Image Processing & Analysis:** ImageJ
- **Software & Documentation:** MS Office, Google Suite

REFERENCES

Available on request.